Pediatric Vision Responsible AI Prototype v0.1

Technical Understanding, User Stories, Acceptance Criteria & UAT Guide

Plain-English + Technical Reference

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Important safety statement
This document describes a synthetic research prototype. Prototype v0.1 is not clinically validated, does not diagnose amblyopia or strabismus, is not a screening tool or medical device, and must not be used for patient care or autonomous referral decisions. Current weights and thresholds are illustrative engineering placeholders only.

1. Executive Summary

In one sentence: Prototype v0.1 is a Python/FastAPI research proof of concept that accepts synthetic pediatric vision-related inputs, checks whether enough information is available, produces a transparent research-only review priority or abstains, and returns the result as FHIR R4-style resources while preserving clinician authority.

Audience	Simple explanation	Technical explanation
Layperson / family	The prototype organizes synthetic information about a child and shows how a future system might flag a case for closer clinician review. It does not diagnose anything.	N/A
Clinician	It demonstrates a clinician-in-the-loop workflow: structured inputs → review priority/abstention → explainable rationale → proposed referral artifact only for workflow testing.	Outputs are non-diagnostic and explicitly labeled research-only.
Engineer / data scientist	It is an API service with Pydantic input validation, a deterministic rule engine, FHIR R4-style resource generation, and pytest coverage for core risk-engine behavior.	FastAPI endpoints expose synthetic demo cases and POST-based assessment.
Research reviewer	It tests workflow, interoperability, transparency, missing-data abstention, and preservation of human oversight before any predictive ML claims.	No clinical performance metrics are claimed in v0.1.

2. What v0.1 Is — and Is Not

Area	What v0.1 does	What v0.1 does NOT do
Clinical purpose	Tests a research workflow concept using synthetic data.	Does not diagnose, screen, treat, or make patient-care decisions.
Decision logic	Uses transparent illustrative weights and priority tiers.	Does not contain a trained or validated ML model.
Missing data	Abstains when fewer than 60% of six research signals are present.	Does not infer missing values or treat missing as “negative.”
Referral workflow	Creates a FHIR-style ServiceRequest only as draft/proposal for high-review-priority cases.	Does not create a real referral order.
Interoperability	Builds FHIR R4-style Patient, Observation, RiskAssessment, ServiceRequest, and Bundle structures.	Does not connect to a live EHR or production FHIR server.
Validation	Includes automated tests for three core engine behaviors.	Does not establish sensitivity, specificity, AUC, clinical utility, or safety in real patients.

3. System Workflow

Plain-English workflow
Synthetic child information → Check whether enough information exists → If too much is missing, stop and say “insufficient data” → Otherwise calculate a transparent research priority → Explain which signals contributed → Package the result in FHIR-style healthcare resources → Keep referral as a clinician-reviewed proposal, never an autonomous order.

Synthetic Record → Pydantic Validation → Completeness Check → Rule Engine → Rationale → FHIR Bundle → Clinician Review

3.1 Technical components

Component	Technology / file	Role
API layer	FastAPI — app/main.py	Provides the home page, demo endpoint, POST assessment endpoint, and auto-generated API documentation.
Input schema	Pydantic ResearchInput — app/main.py	Validates the expected structured fields before assessment.
Decision logic	app/risk_engine.py	Calculates data completeness, abstention, illustrative score, priority tier, and rationale.
FHIR mapping	app/fhir.py	Builds Patient, Observation, RiskAssessment, draft ServiceRequest, and collection Bundle resources.
Synthetic data	data/synthetic_patients.json	Provides three controlled demo cases.
Automated tests	tests/test_risk_engine.py	Checks high priority, missing-data abstention, and routine behavior.
Server runtime	Uvicorn	Runs the FastAPI application locally.

4. Input Data and Current Research Signals

The prototype currently accepts six optional Boolean research signals. Each value can be True, False, or missing (None/null). Missing is intentionally different from False.

Field	Layman meaning	Current illustrative weight
family_history_strabismus_or_amblyopia	Family history of strabismus or amblyopia	2
prematurity_or_low_birth_weight	Prematurity or low birth weight	1
clinician_observed_alignment_concern	A clinician-observed eye alignment concern	3
prior_vision_screen_concern	A prior vision-screen concern	3
caregiver_visual_behavior_concern	A caregiver-reported visual behavior concern	1
developmental_or_neurologic_risk_context	Developmental or neurologic risk context	1

Critical interpretation
The weights above are not medical evidence. They exist only to test software behavior. In any presentation, call them illustrative engineering weights or placeholder logic — never clinical risk weights.

5. Decision Logic in v0.1

5.1 Data completeness and abstention

The system counts how many of the six research signals are known (True or False) and divides that number by six.

Data completeness = known research signals ÷ 6

- If completeness is below 60%, the prototype abstains.
- Abstention output: status = insufficient-data, research_priority = abstain, illustrative_score = None.

- If completeness is 60% or higher, the prototype continues to the illustrative scoring logic.

5.2 Research-only priority tiers

Illustrative score	Output	Meaning in v0.1
0–1	routine-review-priority	Routine research review priority.
2–4	moderate-review-priority	Moderate research review priority.
5+	high-review-priority	High research review priority; triggers a draft/proposed ServiceRequest artifact.

6. FHIR Output — Plain English and Technical Meaning

FHIR resource	Plain-English meaning	How v0.1 uses it
Patient	The synthetic child record	Stores synthetic ID, name, gender, and birth date.
Observation	A structured piece of information about the child	Represents each non-missing research signal as a Boolean observation.
RiskAssessment	The structured research assessment result	Stores the review-priority output, supporting observations, rationale, and disclaimer.
ServiceRequest	A proposed request for a downstream service	Created only for high-review-priority cases, with status=draft and intent=proposal. It is explicitly not a real referral order.
Bundle	A package of related FHIR resources	Returns Patient + Observations + RiskAssessment + optional draft ServiceRequest together.

7. API Endpoints and Demo Behavior

Endpoint	Method	Purpose	Expected audience
/	GET	Simple landing page showing synthetic cases and a link to API docs.	General demo
/demo/{patient_id}	GET	Loads one stored synthetic patient, runs the assessment, and returns assessment + FHIR Bundle.	Research/demo user
/assess	POST	Accepts a ResearchInput JSON payload, validates it, runs the assessment, and returns assessment + FHIR Bundle.	Technical/API user
/docs	GET	FastAPI-generated interactive OpenAPI/Swagger documentation.	Developer/reviewer

8. User Stories

User stories describe desired behavior from the perspective of a user. They are useful because clinicians, researchers, product owners, and engineers can all understand the same requirement.

ID	Persona	User story	Acceptance criterion
US-01	Researcher	As a researcher, I want to run	A valid synthetic case returns

ID	Persona	User story	Acceptance criterion
		a synthetic patient case so that I can observe the end-to-end workflow.	both an assessment object and a FHIR Bundle.
US-02	Clinician reviewer	As a clinician reviewer, I want the system to explain which inputs contributed to the priority so that I can understand the basis of the output.	Triggered signals and rationale are returned for non-abstained cases.
US-03	Safety reviewer	As a safety reviewer, I want the system to abstain when too much data are missing so that it does not create false confidence.	Completeness <60% produces abstain and no illustrative score.
US-04	Integration engineer	As an integration engineer, I want outputs represented in FHIR-style resources so that the workflow can be evaluated for future interoperability.	Bundle contains Patient, available Observations, and RiskAssessment; high-priority cases also contain draft ServiceRequest.
US-05	Clinician reviewer	As a clinician reviewer, I want referral activity to remain a proposal so that the prototype cannot independently order care.	ServiceRequest uses status=draft and intent=proposal and includes a non-clinical disclaimer.
US-06	Developer	As a developer, I want a POST API with schema validation so that structured synthetic inputs can be tested consistently.	/assess accepts ResearchInput-compatible JSON and rejects invalid required structure.
US-07	Research lead	As a research lead, I want the output labeled non-diagnostic so that the prototype is not misrepresented as clinically validated.	Assessment and FHIR output include explicit research-only disclaimers.
US-08	QA tester	As a QA tester, I want known test cases for high, routine, and missing-data behavior so that regressions can be detected.	Existing pytest suite passes all three core behavior tests.

9. Functional Requirements

ID	Requirement	Status
FR-01	The system shall accept structured synthetic patient input through the ResearchInput schema.	Implemented
FR-02	The system shall calculate data completeness across the six research signals.	Implemented
FR-03	The system shall abstain when completeness is below 60%.	Implemented
FR-04	The system shall calculate an illustrative score only when it does not abstain.	Implemented
FR-05	The system shall map the score to routine, moderate, or high research-review priority.	Implemented
FR-06	The system shall return triggered	Implemented

ID	Requirement	Status
	signals and human-readable rationale.	
FR-07	The system shall build FHIR R4-style Patient, Observation, and RiskAssessment resources.	Implemented
FR-08	The system shall create a draft/proposed ServiceRequest only for high-review-priority, non-abstained cases.	Implemented
FR-09	The system shall include explicit non-clinical disclaimers.	Implemented
FR-10	The system shall expose demo and POST assessment endpoints plus interactive API documentation.	Implemented

10. User Acceptance Testing (UAT)

UAT asks a simple question: “Does the prototype behave the way a research user expects?” The cases below combine current demonstrated behavior with recommended acceptance tests for v0.1. Existing automated pytest coverage currently verifies UAT-01, UAT-03, and UAT-04 at the rule-engine level.

ID	Scenario	Test action	Expected result	Coverage
UAT-01	High-priority synthetic case	Use pv-001 / equivalent complete case with family history + alignment concern + prior screen concern = True.	Output is high-review-priority; abstained=False; score present; rationale lists positive signals; Bundle includes draft ServiceRequest.	Core priority behavior automated; Bundle behavior recommended for UAT.
UAT-02	Moderate-priority synthetic case	Use pv-002.	Output is moderate-review-priority; no ServiceRequest is created.	Recommended UAT.
UAT-03	Insufficient-data abstention	Use pv-003 / case with only 2 of 6 signals known.	Output is abstain; status=insufficient-data; score=None; no ServiceRequest.	Abstention automated at rule-engine level.
UAT-04	Routine case	Provide all six research signals as False.	Output is routine-review-priority with illustrative_score=0.	Automated.
UAT-05	Unknown demo patient	Call GET /demo/nonexistent.	API returns HTTP 404 with “Synthetic patient not found.”	Recommended UAT.
UAT-06	Missing required patient_id	POST /assess without patient_id.	Pydantic/FastAPI rejects the request as invalid input.	Recommended UAT.
UAT-07	Observation creation	Provide a mix of known and null signals.	FHIR Observations are created only for non-null research signals.	Recommended UAT.
UAT-08	Clinician authority safeguard	Run a high-priority case.	ServiceRequest has status=draft and intent=proposal and includes explicit “not a clinical referral order” language.	Recommended UAT.
UAT-09	FHIR referential consistency	Run any non-abstained case.	Observation.subject and RiskAssessment.subject reference Patient/{patient_id}; RiskAssessment.basis references generated Observations.	Recommended UAT.
UAT-10	Research disclaimer	Run demo/API result.	Assessment and FHIR	Recommended UAT.

ID	Scenario	Test action	Expected result	Coverage
	visibility		RiskAssessment include research-only/non-diagnostic disclaimer.	

11. Current Automated Test Coverage

Test	What it verifies
test_high_priority_case	A complete record with several positive signals produces high-review-priority and does not abstain.
test_abstention_for_missing_data	A sparse record produces abstain, abstained=True, and illustrative_score=None.
test_routine_case	A complete record with all six signals False produces routine-review-priority and score 0.

Important QA distinction
Automated unit tests verify code behavior. UAT verifies whether the end-to-end behavior is acceptable to intended research users. A mature v0.2 should add API/FHIR integration tests in addition to the current rule-engine tests.

12. How to Explain the Three Demo Cases

Synthetic case	What happens	How to explain it
pv-001 — Avery	Complete data with three positive higher-weight signals → high-review-priority → draft ServiceRequest proposal.	"This demonstrates a case where the workflow asks for closer human review and creates a proposed downstream artifact, but it still does not make a clinical referral."
pv-002 — Jordan	Complete data with two lower-weight positive signals → moderate-review-priority → no ServiceRequest.	"This demonstrates a middle-priority research case where the system provides explainable support but does not create a referral proposal."
pv-003 — Riley	Only 2 of 6 research signals known → about 33% completeness → abstain.	"This demonstrates the safety behavior: the prototype refuses to create a score when too much information is missing."

13. Short Demo Script

13.1 30-second layperson explanation

"This prototype uses completely synthetic child records to test how a future clinician-supervised AI workflow might operate. It first checks whether enough information is available. If too much is missing, it abstains. If enough data are available, it produces a transparent research-only review priority and explains why. The output is packaged in FHIR-style healthcare data structures, but the system never diagnoses a child or makes a real referral decision.**"**

13.2 60-second technical explanation

"Prototype v0.1 is a FastAPI service with Pydantic input validation, a deterministic rule engine, and FHIR R4-style output mapping. The engine evaluates six synthetic research signals, calculates data completeness, and abstains below a 60% completeness threshold. Otherwise it applies illustrative, non-clinical weights to produce routine, moderate, or high research-review priority with explicit rationale. The FHIR layer creates Patient, Observation, and RiskAssessment resources, plus a draft/proposed ServiceRequest only for high-priority workflow testing. Current pytest coverage verifies high-priority, routine, and missing-data abstention behavior.**"**

14. Glossary

Term	Simple meaning	Technical meaning
API	A controlled way for software systems to communicate.	HTTP endpoints that accept requests and return structured responses.
FastAPI	The web framework running the prototype.	Python ASGI framework used to define GET/POST routes and OpenAPI docs.
Pydantic	Checks that incoming data have the expected shape.	Python schema validation/serialization library used by FastAPI.
FHIR	A healthcare data-sharing standard.	HL7 Fast Healthcare Interoperability Resources.
Abstention	The system chooses not to give an answer.	A safety behavior triggered in v0.1 when data completeness is <60%.
Explainability	Showing why the system produced an output.	Triggered signals and rationale are returned instead of a score alone.
Synthetic data	Artificial example data, not real patient data.	Controlled test records used to evaluate workflow behavior without clinical claims.
User story	A requirement written from a user's perspective.	Format: As a [persona], I want [capability], so that [benefit].
Acceptance criterion	The condition that must be true for a user story to be accepted.	Observable, testable behavior tied to a requirement.
UAT	A final check from the user/business/research perspective.	User Acceptance Testing validates that end-to-end behavior meets intended needs.
Unit test	A code-level check for one piece of logic.	Automated pytest test of isolated behavior, such as abstention.

15. Known v0.1 Limitations and Documentation Next Steps

- All patient data are synthetic.
- The six research signals, weights, 60% abstention threshold, and score tiers are illustrative engineering placeholders.
- No trained machine-learning model is present in v0.1.
- No live EHR/FHIR server integration is present.
- No clinical validation or diagnostic performance claims are supported.
- No fairness, subgroup, audit-log, longitudinal referral, or follow-up evaluation is implemented yet.
- Current automated tests focus on the rule engine; API/FHIR integration tests should be added in a later version.

Recommended next documentation package for v0.2: Research protocol, expanded data dictionary, traceability matrix (requirement → user story → test case), API/FHIR integration test plan, synthetic cohort specification, and change log.

Appendix A. Source-of-Truth Files for v0.1

File	What it defines
app/main.py	FastAPI application, Pydantic ResearchInput schema, /, /demo/{patient_id}, /assess endpoints.
app/risk_engine.py	Six signals, illustrative weights, 60% completeness threshold, priority tiers, rationale, disclaimer.
app/fhir.py	FHIR-style Patient, Observation, RiskAssessment, ServiceRequest, and Bundle generation.
data/synthetic_patients.json	Three demo patients: pv-001, pv-002, pv-003.
tests/test_risk_engine.py	Three current automated tests.

File	What it defines
docs/research_scope.md	Purpose, explicit non-claims, v0.1 research questions, and next validation layers.